T-Shirt Cannon Control PCB Formal Requirements

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This document contains the formal requiremets for the power distribution board for the Rocker Robotics Space Elevator.

1. It must use a raspberry pi pico for the main control.
2. It must take a 4s LIPO battery as an input.
3. It must have power breakout for four motors with full battery power.
4. It must have a 5v output for the control board.
5. It must measure the ammount of power that the robot is drawing.
6. It must communicate the power draw with the main control board over I2C.
7. It must have a relay to cut power to the robot.
8. It must have a switch that will act as an emergency stop/power button.
9. It must cut power if a signal is recived from the control board.
10. It must communicate current battery voltage to the main control board.
11. It must handle regenerative breaking from the VESC ESCs.
12. It must have an input power LED.
13. It must have a 5v output power LED.